Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/24/2007
PI Name:	Delp, Michael Ph.D.		
Project Title:	Adaptations of Cerebral Arteries to Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	08/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI moved to West Va. U from TX A&M in Sept 2005; grant transferred to WVU; will be new grant NNA06CB42G (per A. Chu 7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC66G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	These studies will address the relatively unexplored question of the effects of simulated microgravity on vascular smooth muscle and endothelial cell function in resistance arteries isolated from splanchnic, renal and cerebral tissues. Alterations in vascular function induced by microgravity are particularly relevant to the problems of orthostatic intolerance experienced by astronauts upon re-entry into the earth's gravitational field. Previous research has demonstrated that reductions in blood volume and altered baroreceptor reflex control of the cardiovascular system contribute to, but do not fully account for, the changes in orthostatic tolerance. The problem of altered resistance vessel reactivity and morphology will be studied using a ground-based model to simulate a microgravity environment, i.e., the tail-suspended, hindlimb-unloaded rat. The impact of simulated microgravity on maximal responsiveness and sensitivity of resistance arteries to a variety of vasoconstrictor and vasodilator agents, each acting through different mechanisms, will be determined. This work will provide potentially important information about the mechanism(s) underlying the orthostatic intolerance of astronauts returning to Earth. Such information would be useful in the development of more effective countermeasures. The impact of simulated microgravity on maximal responsiveness and sensitivity of resistance arteries to a variety of vasoconstrictor and vasodilator agents, each acting through different mechanisms, will be determined. This work will provide potentially important information about the mechanism(s) underlying the orthostatic intolerance of astronauts returning to Earth. Such information would be useful in the development of more effective countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments will expand our understanding of how prolonged fluid shifts, such as in a microgravity environment or during bedrest, affect the cardiovascular system. The detailed study of the resulting vascular adaptations triggered by simulated microgravity will yield essential information on the basic physiological responses of individual blood vessels involved in blood flow and pressure regulation. This information will also support the development of treatments or countermeasures to improve crew health and performance following their return to a gravitational environment.
Task Progress:	I. Increases in myogenic, shear stress and agonist-induced vasoconstriction in the middle cerebral artery (MCA) with head-down tail-suspension (HDT) in rats are the result of alterations in the release of the endothelium-derived relaxing factor, nitric oxide (NO). Myogenic, shear stress, and KCl-induced vasoconstriction will be tested in the MCA from control and HDT rats with an intact and removed endothelial cell layer and in the presence and absence of the NO synthase inhibitor NG-nitro-L-arginine methyl ester (L-NAME). These studies have been completed and the results have been published in the American Journal of Physiology: Heart and Circulatory Physiology (Wilkerson, MK, LA Lesniewski, EM Golding, RM Bryan, A Amin, E Wilson, and MD Delp. Simulated microgravity enhances cerebral artery vasoconstriction and vascular resistance through an endothelial nitric oxide mechanism. Am. J. Physiol. Heart Circ. Physiol. 288: H1652-H1661, 2005). The results demonstrate that simulated microgravity enhances basal tone and vasoconstrictor responsiveness of cerebral arteries, therefore potentially limiting cerebral perfusion during period of altered gravitational stress (e.g., during standing after return from a weightless environment). When applied to the human condition, these results suggest that alterations in cerebral autoregulation associated with microgravity or prolonged bedrest may be the result of enhanced vasoconstrictor tone of cerebral arteries. The significance of these data are demonstrated in their use (in April 2005) by Douglas C. Eaton, Ph.D., President of the American Physiological Society, in a letter urging US Senators to increase support for NASA peer-reviewed research into the health risks of long-term space flight . II. The mechanism through which the endothelium mediates vasodilation of MCAs is altered by HDT, so that there is less vasodilation through the nitric oxide synthase pathway and more vasodilation through the release of endothelium-derived hyperpolarizing factor (EDHF). Vasodilation induced by 2-methylthio-ATP (2MeS-ATP) and bradykinin, two endothelium-dependent vasodilators, will be determined in the presence of the NO synthase inhibitor L-NAME, the cyclooxygenase inhibitor indomethacin, and the Ca2+-activated K+ channel inhibitors charybdotoxin and apamin. Experiments to determine the effects of simulated microgravity on endothelium-dependent and endothelium-independent vasodilation of cerebral arteries have been completed. The results demonstrate that endothelium-dependent vasodilation to 2MeS-ATP and bradykinin is diminished by simulated microgravity. However, endothelium-independent vasodilation through sodium nitroprusside is not altered. Currently experiments are in progress to determine which signaling pathway is altered by simulated microgravity to account for the impaired endothelium-dependent vasodilation. These mechanistic studies were scheduled to be completed at the end of year 2 and beginning of year 3 and will be published when completed with the data demonstrating simulated microgravity impairs endothelium-dependent vasodilation in cerebral arteries. III. The smooth muscle hypertrophy and the corresponding increase in MCA medial thickness induced by HDT will diminish the passive pressure-diameter relationship and increase the stiffness of MCAs. The structural behavior and material properties of the MCA from control and HDT rats will be compared to determine whether structural vascular alterations affect the stiffness of these arteries, and thus, how readily the arteries will vasodilate during smooth muscle cell relaxation. No progress has been made toward this aim. These studies are scheduled to be conducted during year 3 and will determine whether changes in the mechanical properties of the cerebral arteries (i.e., increased stiffness) contributes to decrements in cerebral perfusion associated with simulated microgravity (see results of Specific Aim #4 below). IV. The net effect of HDT-induced alterations in cerebral resistance artery vasoconstrictor responsiveness, vasodilator responsiveness, or mechanical properties is a diminished cerebral perfusion and elevated cerebral vascular resistance. Total cerebral perfusion, brain blood flow distribution, and total and regional cerebral vascular resistance will be determined in control and HDT rats during horizontal (0o tilt) and 75o head-up tilt. These studies have been completed and the results have been published in the American Journal of Physiology: Heart and Circulatory Physiology. The results demonstrate that following a 2 wk period of simulated microgravity, cerebral perfusion is lower and vascular resistance is higher in several regions of the brain when the animals are normally standing in the horizontal position. During head-up tilt, the HDT rats become hypotensive. However, despite this hypotension, cerebral vascular resistance is higher and perfusion is lower than that of control animals in several regions

	of the brain. Therefore, these data suggest that spaceflight and bedrest-induced deficits in locomotion, balance and orthostatic tolerance may be related to regional reductions in perfusion of key regulatory areas within the brain.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Wilkerson MK, Lesniewski LA, Golding EM, Bryan RM Jr, Amin A, Wilson E, Delp MD. "Simulated microgravity enhances cerebral artery vasoconstriction and vascular resistance through endothelial nitric oxide mechanism." Am J Physiol Heart Circ Physiol. 2005 Apr;288(4):H1652-61. Epub 2004 Dec 2. PMID: 15576439 , Apr-2005